

Standing order of the final project in the course

“Data Security and Cryptology”

A project suggests an implementation of a data security mechanism in applications like:

- Secure emails exchange,
- Secure access to documents editing,
- Secure payment.

The course supervisors himself determines the project topic for each group.

The system has to include:

- An implementation of one of the symmetric cryptographic algorithms in one of the operation modes. The algorithm is intended to encrypt and decrypt the data.
- An implementation of one of the Public-key cryptographic algorithms. The algorithm is intended to create a secure communication channel.
- An implementation of one the Digital Signature schemes.
- Implementation of the solution in any programming language like Python, C++, etc.
- It is not allowed to use toolkits like “import cryptography” of Python or "include" of C. An implementation should be self-provided. It is possible to use such an extension to the calculation of Hash Functions and Key generation.
- GUI is not a necessary attribute, but a possibility to evaluate and to control the input and the output has to be.
- The project submission deliverables:
 - A two pages report in English summarizing the process of the project study, the project flow, the obtained results, and conclusions
 - The source code files
 - A PowerPoint presentation in English.
- The project is presented in a frontal manner in a classroom within the overall 30 minutes. The recommended size of a group is 2 or 3 students (not 4). The students are requested to explain not only the project itself but also the theoretical background connected to the subject, especially the reasoning of the proposed solution selection and the result evaluation.